BDV Syndrome (Blakemore–Durmaz–Vasileiou Syndrome): Comprehensive Disease Characterization

Disease: BDV Syndrome | **MONDO ID:** MONDO:0859150 | **Gene:** *CPE* (Carboxypeptidase E) | **Inheritance:** Autosomal recessive | **Category:** Mendelian

Summary

BDV syndrome (Blakemore–Durmaz–Vasileiou syndrome; MONDO:0859150) is an ultra-rare autosomal-recessive Mendelian disorder caused by biallelic loss-of-function (LoF) variants in *CPE*, the gene encoding carboxypeptidase E. CPE is the exopeptidase that performs the final C-terminal trimming step in prohormone/proneuropeptide maturation, removing C-terminal basic residues (Lys/Arg) left behind by the prohormone convertases. It also doubles as a secretory-granule sorting receptor and as a neurotrophic/neuroprotective factor (Neurotrophic factor-α1, NF-α1). When CPE function is abolished, bioactive peptides across many neuroendocrine systems — insulin, POMC-derived α-MSH, and appetite/glucose/reproduction-regulating peptides — fail to mature, while inactive intermediates accumulate.

The clinical consequence is a near-fully-penetrant, **Prader-Willi-like triad**: childhood-onset severe hyperphagic obesity, intellectual disability/neurodevelopmental delay, and hypogonadotropic hypogonadism. Abnormal glucose homeostasis and type 2 diabetes are frequent, and elevated plasma proinsulin is a mechanistically grounded candidate biomarker. The disorder has been reported in fewer than ~15 affected individuals from roughly six consanguineous families worldwide (British/European, Turkish, Syrian, Egyptian, Pakistani ancestries). Diagnosis is made by exome or genome sequencing, generally after Prader-Willi syndrome (PWS) has been excluded, because the phenotype is clinically mistaken for PWS.

There is **no approved disease-specific therapy**; management is supportive and multidisciplinary (dietary/hyperphagia management, diabetes treatment, sex-hormone replacement, developmental/educational support). The MC4R agonist setmelanotide — approved for other leptin-melanocortin-pathway monogenic obesities (POMC, PCSK1, LEPR

deficiency; Bardet-Biedl syndrome) — is a biologically rational but unvalidated candidate, since CPE acts upstream of α-MSH. Two mouse models, *Cpe*^fat/fat^ and *Cpe*-knockout, faithfully recapitulate the human metabolic, reproductive, and cognitive phenotypes and reveal CPE's role in hippocampal CA3 neuron survival.

Key Findings

Finding 1 — BDV syndrome is caused by biallelic loss-of-function *CPE* variants (autosomal recessive)

Three independent families/cohorts established that biallelic LoF *CPE* variants co-segregate with a consistent phenotype of morbid/childhood-onset obesity, intellectual disability/neurodevelopmental delay, and hypogonadotropic hypogonadism. The syndrome was formally named and standardized against Human Phenotype Ontology terms by Durmaz and colleagues: *"Comparison of clinical features with previously described cases and standardization according to the Human Phenotype Ontology terms indicated a recognizable clinical phenotype, which we termed Blakemore-Durmaz-Vasileiou (BDV) syndrome"* (PMID: 34383079).

The first human case was a morbidly obese female from a consanguineous family homozygous for a truncating mutation (c.76_98del; p.E26RfsX68), with *"no CPE expression in whole blood-derived RNA from the proband, consistent with nonsense-mediated decay"* (PMID: 26120850). A second independent family carried a homozygous nonsense c.405C>A (p.Y135*) mutation in three affected siblings (PMID: 32936766). Additional pathogenic alleles include c.361C>T p.(Arg121*) and c.994del p.(Ser333Alafs*22) (Bosch 2021, PMID: 34383079). All reported families were consanguineous, and proband RNA studies support nonsense-mediated decay as the molecular mechanism (true null alleles).

Reported pathogenic *CPE* variants

Variant (cDNA)	Protein	Type	Zygosity	Source
c.76_98del	p.E26RfsX68	Frameshift (truncating)	Homozygous	PMID: 26120850
c.405C>A	p.Y135*	Nonsense	Homozygous	PMID: 32936766
c.361C>T	p.(Arg121*)	Nonsense (recurrent/ founder)	Homozygous	PMID: 34383079
c.994del	p.(Ser333Alafs*22)	Frameshift	Homozygous	PMID: 34383079

Finding 2 — Mechanism: CPE loss abolishes C-terminal prohormone processing, accumulating inactive intermediates and depleting mature bioactive peptides

CPE is *"an exopeptidase that removes C-terminal basic amino acids from intermediates to produce bioactive peptides. The mutation renders the enzyme inactive and unstable. The absence of Cpe activity in these mutants leads to abnormal processing of many peptides, with elevated levels of intermediates and greatly reduced levels of the mature peptides"* (PMID: 23442571). It is *"a prohormone-processing enzyme and peripheral membrane protein of endocrine/ neuroendocrine secretory granules"* that also functions as a sorting receptor and co-aggregates with granule content proteins at acidic pH (~5.5–5.8) (PMID: 9813022).

Because CPE *"is a peptide processing enzyme, involved in cleaving numerous peptide precursors, including neuropeptides and hormones involved in appetite control and glucose metabolism"* (PMID: 26120850), its loss directly links the molecular defect to the appetite, glucose, reproductive, and cognitive phenotypes observed in patients. This is a **loss-of-function** mechanism affecting the terminal, convergent step of neuroendocrine peptide maturation — explaining the pleiotropic, multi-system presentation.

Finding 3 — Mouse models (*Cpe*^fat/fat^ and *Cpe*-knockout) recapitulate the human disease and reveal a neuroprotective role

Cpe^fat/fat^ mice (Ser202Pro point mutation) *"develop obesity, diabetes and infertility in adulthood"* (PMID: 23442571), plus age-dependent depressive-like (~60 days) and anxiety-like (>90 days) behaviours. *Cpe*-knockout mice show memory consolidation deficits, absent long-

term potentiation (LTP), and *"marked degeneration specifically of the pyramidal neurons in the hippocampal CA3 region which normally expresses high levels of CPE"* from 4 weeks of age (PMID: 18570185). Neuron-specific Camk2a-Cre conditional knockout reproduces learning/memory deficits and neurodegeneration (PMID: 37100779).

Genetic background strongly modifies severity: *"On the HRS/J (HRS) inbred strain background, mice homozygous for the fat mutation exhibit early onset hyperinsulinemia followed by postpubertal moderate obesity without hyperglycemia. In contrast, on the C57BLKS/J (BKS) genetic background, fat/fat mice become severely obese, hyperinsulinemic, and hyperglycemic"* (PMID: 15870393). This demonstrates that genetic modifiers can shape the obesity/diabetes phenotype and provides a model-based rationale for variable expressivity. Human patient phenotypes (obesity, intellectual disability, abnormal glucose homeostasis, hypogonadotropic hypogonadism) explicitly recapitulate the fat/fat and Cpe-KO models.

Finding 4 — CPE loss disrupts proinsulin processing, elevating plasma proinsulin (candidate biomarker)

Human CPE-mutation carriers *"have elevated plasma proinsulin and develop severe obesity and hyperglycemia"* (PMID: 37364047). A pancreatic β-cell-specific knockout (βCpeKO; Cpefl/fl × Ins1Cre/+) model showed that these mice *"lack mature insulin granules and have elevated proinsulin in plasma"* (PMID: 37364047), though glucose-/KCl-stimulated secretion remained intact and spontaneous diabetes did not develop on high-fat diet. However, β-cells showed reduced mitochondrial membrane potential, increased reactive oxygen species, reduced MafA, elevated Aldh1a3 under high-glucose stress, and accelerated hyperglycemia after low-dose streptozotocin — indicating increased β-cell vulnerability rather than fixed failure. CPE's role in proinsulin processing is corroborated in human islets, where *"Laser capture microdissection followed by mass spectrometry revealed reductions in the proinsulin processing enzymes prohormone convertase 1/3 (PC1/3) and carboxypeptidase E (CPE) in T1D donors"* (PMID: 31442418).

Diagnostic implication: an elevated **proinsulin-to-insulin ratio** is a mechanistically grounded biochemical signature of CPE deficiency and a candidate screening/diagnostic biomarker worth prospective evaluation.

Finding 5 — CPE's oncogenic splice variant (CPE-ΔN) is a distinct biology, not the BDV mechanism

An N-terminal truncated splice variant *"(CPE-ΔN) has been cloned and found to be highly expressed in malignant tumors and cell lines derived from prostate, breast, liver and lung cancers and gliomas"* (PMID: 35988818). CPE-ΔN acts in the nucleus with HDACs to drive metastasis — e.g., it *"has recently been recognized as an independent predictor for the recurrence and metastasis of lung adenocarcinoma"* by repressing E-cadherin via stabilizing the Snail-HDAC complex (PMID: 32266100). This is a **gain-of-function, isoform-specific** oncogenic role distinct from the recessive LoF that causes BDV syndrome. It is important context for interpreting *CPE* literature: elevated CPE/CPE-ΔN in cancer is not relevant to the loss-of-function disease, and the two should not be conflated.

Finding 6 — Phenotype spectrum: a near-fully-penetrant triad with abnormal glucose homeostasis

Across the ~8–11 clearly reported affected individuals (Alsters 2015 n=1; Durmaz 2021 n=3; Bosch 2021 n=4 from 3 families plus 4 previously reported from 2 families), essentially all exhibit the core triad of (1) childhood-onset morbid obesity with hyperphagia, (2) intellectual disability/neurodevelopmental delay, and (3) hypogonadotropic hypogonadism, described as *"core clinical features including morbid obesity, neurodevelopmental delay, and hypogonadotropic hypogonadism"* (PMID: 34383079). An independent family confirmed *"three siblings with obesity, intellectual disability and hypogonadotropic hypogonadism"* (PMID: 32936766). The clinical overlap with Prader-Willi syndrome drives the diagnostic pathway: *"After excluding Prader-Willi syndrome (PWS), exome sequencing was performed in both Syrian siblings"* (PMID: 34383079).

Phenotype → HPO term mapping

Phenotype	HPO term	Frequency	Onset
Childhood-onset morbid obesity	HP:0025499 (Childhood-onset obesity)	~Universal	Childhood
Hyperphagia	HP:0002591 (Polyphagia)	Frequent	Childhood
Intellectual disability	HP:0001249	~Universal	Childhood
Global developmental delay	HP:0001263	Frequent	Infancy/ childhood
Hypogonadotropic hypogonadism	HP:0000044	~Universal	Peripubertal
Type 2 diabetes / abnormal glucose homeostasis	HP:0005978 / HP:0011014	Variable/ frequent	Childhood–adult
Behavioral abnormality (model-predicted)	HP:0000708	Inferred (mouse)	—
Memory impairment (model-predicted)	HP:0002354	Inferred (mouse)	—

Finding 7 — Identifiers, inheritance, and ultra-rare epidemiology

Gene identifiers: *CPE* (carboxypeptidase E; aka CPH, NF-α1): HGNC:2303; NCBI Gene 1363; Ensembl ENSG00000109472; cytoband 4q32.3; OMIM gene *114855; UniProt P16870 (CBPE_HUMAN). **Disease:** MONDO:0859150; synonyms "Blakemore-Durmaz-Vasileiou syndrome," "CPE deficiency," "carboxypeptidase E deficiency."

Inheritance and epidemiology: autosomal recessive (biallelic LoF). The disorder is ultra-rare — fewer than ~15 affected individuals across ~6 families to date. Durmaz/Bosch describe *"4 affected individuals from 3 unrelated consanguineous families, 2 siblings of Syrian, 1 of Egyptian, and 1 of Pakistani descent, all harboring novel homozygous CPE LoF variants"* (PMID: 34383079). A recurrent/possible founder allele is noted: *"Three individuals from 2 families shared the same CPE homozygous truncating variant c.361C > T, p.(Arg121*)"* (PMID: 34383079). The first reported case involved *"a morbidly obese female from a consanguineous family… homozygosity for a truncating mutation of the CPE gene"* (PMID: 26120850). Biallelic CPE LoF is essentially absent in gnomAD (no reported homozygous healthy LoF carriers), consistent with a recessive, highly penetrant disorder. No formal population prevalence/incidence estimate exists.

Finding 8 — Treatment is supportive; setmelanotide is a mechanistically rational candidate

No disease-specific or curative therapy exists. Management is supportive and organ-directed: dietary/caloric restriction and hyperphagia management, type 2 diabetes treatment, sex-hormone replacement for hypogonadotropic hypogonadism (induction of puberty/fertility support), and developmental/educational/behavioral support for intellectual disability. The MC4R agonist setmelanotide is approved for upstream leptin-melanocortin-pathway defects — *"those with LEPR, PCSK1, or POMC deficiency can now be treated with the MC4R agonist setmelanotide"* (PMID: 39777073) — and *"compensates for upstream gene defects in the brain leptin-melanocortin pathway and reduces hyperphagia and obesity in selected monogenic obesity forms and Bardet-Biedl syndrome (BBS)"* (PMID: 40903014). Because CPE acts upstream (producing α-MSH), setmelanotide is biologically plausible but not yet approved/validated for CPE deficiency. Bariatric surgery has limited durable benefit in hyperphagic syndromic obesity: *"Benefits of bariatric surgery may be less durable"* in such patients (PMID: 35416416).

Finding 9 — CPE/NF-α1 is a neurotrophic/neuroprotective factor enriched in hippocampal CA3

CPE is also known as Neurotrophic factor-α1 (NF-α1). A dedicated review states that *"Families carrying null and homozygous non-sense mutations of the NF-α1/CPE gene share common clinical features including childhood onset obesity, type 2 diabetes, impaired intellectual abilities and hypogonadotrophic hypogonadism"* and that *"NF-α1/CPE expression is enriched in CA3 neurons of the hippocampus"* (PMID: 35711734). CPE exerts enzyme-independent neuroprotection — *"overexpression of CPE in cultured hippocampal neurons protected them against H(2)O(2) oxidative-stress induced cell death"* (PMID: 18570185). This CA3-enriched, neuroprotective role provides the mechanistic basis linking CPE loss to intellectual disability and the model-predicted depression/anxiety/memory phenotypes.

Mechanistic Model / Interpretation

Ordered causal chain (from initiating lesion to clinical manifestation)

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1. Biallelic LoF variant in CPE (nonsense/frameshift, e.g., p.Arg121*, p.Y135*)
        │  leads to →
2. Nonsense-mediated decay of CPE mRNA → complete absence of CPE protein
        │  results in →
3. Loss of the terminal C-terminal-basic-residue trimming step of prohormone
   maturation (and loss of CPE's granule-sorting + neurotrophic functions)
        │  results in →
4. Accumulation of inactive peptide intermediates + depletion of mature
   bioactive peptides across multiple neuroendocrine systems
        │
        ├── BRANCH A (hypothalamic leptin–melanocortin) → deficient α-MSH
        │        → impaired MC4R satiety signalling [inferred]
        │        → hyperphagia → childhood-onset severe obesity (HP:0025499)
        │
        ├── BRANCH B (pancreatic β-cell) → impaired proinsulin C-terminal
        │        trimming → elevated plasma proinsulin, defective mature
        │        insulin granules → β-cell stress/vulnerability
        │        → abnormal glucose homeostasis / type 2 diabetes (HP:0005978)
        │
        ├── BRANCH C (hypothalamic–pituitary–gonadal) → deficient processing
        │        of reproductive neuropeptides → hypogonadotropic
        │        hypogonadism (HP:0000044)
        │
        └── BRANCH D (hippocampus/CNS) → loss of NF-α1 neurotrophic/neuro-
                 protective support, CA3 pyramidal neuron vulnerability
                 → intellectual disability (HP:0001249); (mouse-inferred:
                 memory deficits, depression/anxiety-like behaviour)
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Steps 1–4 are demonstrated in humans (genetics, absent expression) and mice (peptide processing). Branch A's MC4R-satiety link is inferred by analogy to the leptin-melanocortin pathway rather than directly demonstrated in CPE patients. Branch D's cognitive mechanism is strongly supported in mouse models and inferred for the human intellectual disability.

Upstream vs downstream, cell types, and processes

- **Upstream (initiating):** the *CPE* null genotype and absent enzyme/sorting/neurotrophic activity.
- **Downstream (effector):** tissue-specific peptide deficiencies producing each clinical branch.

- **Cell types involved (CL terms):** pancreatic β-cell (CL:0000169), hypothalamic neurons/POMC neurons, hippocampal CA3 pyramidal neurons (CL:0000598, pyramidal neuron), photoreceptors (mouse retinal phenotype, PMID: 16219026).
- **Subcellular compartment (GO CC):** secretory granule (GO:0030141), regulated secretory pathway.
- **Biological processes (GO BP):** peptide hormone processing (GO:0016486), neuropeptide signaling (GO:0007218), insulin processing (GO:0030070), regulation of feeding behavior (GO:0033059), neuron development/survival.
- **Anatomy (UBERON):** hypothalamus (UBERON:0001898), pituitary gland (UBERON:0000007), pancreatic islet (UBERON:0000006), hippocampus/CA3 (UBERON:0003881), retina (UBERON:0000966).

Section-by-section synthesis against the research template

- **Disease information:** autosomal-recessive CPE deficiency; identifiers as in Finding 7; information is derived from aggregated disease-level case reports and model-organism studies (not large EHR cohorts).
- **Etiology:** monogenic (biallelic *CPE* LoF); principal non-genetic "risk factor" is **consanguinity**, which raises homozygosity; no environmental cause. Genetic modifiers (mouse strain background) influence severity.
- **Phenotypes:** see Finding 6 table.
- **Genetic/molecular:** truncating nonsense/frameshift variants → NMD → null; LoF consequence; CPE-ΔN oncogenic isoform is a separate biology (Finding 5).
- **Environmental/infectious:** none implicated. (Note: "BDV" also abbreviates Borna disease virus and Border disease virus/Bovine viral diarrhea in the literature — these are **unrelated** to BDV *syndrome* and were correctly excluded.)
- **Mechanism/pathophysiology:** causal chain above.
- **Anatomy:** hypothalamus, pituitary, pancreatic islet, hippocampus; multi-system (nervous, endocrine, reproductive).
- **Temporal development:** congenital genetic basis; obesity/developmental delay manifest in childhood; hypogonadism peripubertal; chronic lifelong course.
- **Inheritance/population:** AR; high penetrance; enriched in consanguineous families (Finding 7).
- **Diagnostics:** exome/genome sequencing after PWS exclusion; biochemical clue = elevated proinsulin / proinsulin:insulin ratio.

- **Prognosis:** chronic, non-fatal in short term but with obesity/diabetes morbidity and intellectual disability; natural history poorly quantified.
- **Treatment/prevention:** supportive; genetic counseling and cascade/carrier testing in consanguineous families; setmelanotide a rational candidate.
- **Other species/models:** mouse *Cpe*^fat/fat^ and *Cpe*-KO (Finding 3); CPE is highly conserved.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
34383079	*BDV Syndrome: An Emerging Syndrome…Resembling Prader-Willi*	Human clinical	Names syndrome; triad; consanguinity; recurrent p.Arg121* (F1, F6, F7)
26120850	*Truncating Homozygous CPE Mutation in Morbidly Obese Female*	Human clinical	First case; NMD; substrate/appetite link (F1, F2, F7)
32936766	*New Obesity Syndrome…CPE in Three Siblings*	Human clinical	Independent family; homozygous p.Y135* (F1, F6)
23442571	*Anxiety-like behaviours in Cpe(fat/fat) mice*	Model organism	CPE enzymology; fat/fat metabolic + behavioral phenotype (F2, F3)
18570185	*Absence of CPE → hippocampal degeneration & memory deficits*	Model organism	CA3 degeneration; neuroprotection (F3, F9)
15870393	*Genetic modifiers interact with Cpe(fat)*	Model organism	Strain-dependent severity (F3)
9813022	*CPE co-aggregates with granule proteins at acidic pH*	In vitro	Granule sorting/biology (F2)
37364047	*Deletion of CPE in β-cells disrupts proinsulin processing*	Model organism + human	Elevated proinsulin biomarker; β-cell vulnerability (F4)
31442418	*Proinsulin processing abnormalities in T1D islets*	Human	CPE role in human proinsulin processing (F4)
35988818	*CPE and its splice variants in cancer*	Review	Oncogenic CPE-ΔN (distinct biology) (F5)
32266100	*CPEΔN represses E-cadherin via Snail-HDAC*	In vitro/human	Gain-of-function metastasis role (F5)
35711734	*NF-α1/CPE in Neuroprotection and Depression*	Review	Four-feature phenotype; CA3 enrichment (F9)
37100779	*CPE conditional KO: learning/memory deficits*	Model organism	Neuron-specific cognitive phenotype (F9)

PMID	Title (abbrev.)	Evidence type	Supports
39777073	*Precision medicine in rare genetic obesity*	Review	Setmelanotide for upstream pathway defects (F8)
40903014	*Setmelanotide impact in Bardet-Biedl*	Clinical	Setmelanotide mechanism (F8)
35416416	*Bariatric outcomes in PWS/ hyperphagic disorders*	Review	Limited durability of surgery (F8)
16219026	*CPE required for photoreceptor synaptic transmission*	Model organism	Additional neural role (context)

Note on "BDV" nomenclature: Several retrieved papers (e.g., PMID: 24170181, PMID: 22290118, PMID: 37898798) concern **Borna disease virus** or **Border/Bovine viral diarrhea virus** — unrelated to BDV *syndrome*. These were correctly identified as homonyms and excluded from the disease model.

Limitations and Knowledge Gaps

1. **Very small patient base (<15 individuals).** All phenotype frequencies, penetrance, and expressivity estimates are derived from case reports of consanguineous families; ascertainment bias toward severe presentations is likely. No formal prevalence/incidence exists.
2. **No natural-history data.** Disease course, progression rate, survival, long-term complications, and quality-of-life outcomes are not systematically characterized in humans.
3. **Biomarker unvalidated prospectively.** Elevated proinsulin / proinsulin:insulin ratio is mechanistically grounded and observed, but not established as a validated diagnostic/ screening test with sensitivity/specificity data.
4. **Human cognitive/behavioral mechanism inferred from mice.** Depression/anxiety-like and memory phenotypes are robustly demonstrated in *Cpe* models but not formally characterized in patients; the human intellectual disability mechanism is inferred from CA3/neurotrophic biology.

5. **Therapeutics untested.** Setmelanotide is biologically rational but has no reported use in CPE deficiency; its efficacy depends on intact downstream MC4R signalling and sufficient α-MSH-independent response.
6. **Genotype–phenotype correlations unknown.** With only truncating null alleles reported, the phenotypic consequences of hypomorphic/missense *CPE* variants, and the role of modifier loci (as seen in mice), are undefined in humans.
7. **Branch A (α-MSH/MC4R) not directly demonstrated** in CPE patients — the satiety-signalling step is inferred from pathway analogy.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for biallelic *CPE* LoF to aggregate phenotype frequencies, natural history, penetrance, expressivity, and QoL data across the dispersed consanguineous families.
2. **Prospectively validate plasma proinsulin and the proinsulin:insulin ratio** as a diagnostic/screening biomarker in suspected cases (and in PWS-like, PWS-excluded obesity), with defined cutoffs and test performance.
3. **Pilot setmelanotide (MC4R agonist) in CPE deficiency** — a biologically rational n-of-few trial measuring hyperphagia, weight, and metabolic endpoints; include biomarker response (α-MSH-independent pathway intact distal to CPE).
4. **Deep neurocognitive/psychiatric phenotyping** of patients (memory, mood, anxiety) to test whether mouse-predicted CNS features (CA3-linked memory deficits, depression/anxiety) manifest in humans, informing surveillance.
5. **Functional characterization of candidate hypomorphic/missense *CPE* variants** (enzyme activity, secretory-granule sorting, stability) to define genotype–phenotype relationships and resolve VUS.
6. **Modifier-gene mapping**, leveraging the strong strain-background effect in *Cpe*^fat/fat^ mice, to identify loci that could explain variable human severity and inform precision management.
7. **Cascade carrier testing and genetic counseling** in affected consanguineous families, with offer of prenatal/preimplantation testing, as a concrete preventive action given the recessive, high-penetrance inheritance.

Report compiled from 9 confirmed findings across 5 investigation iterations and 35 reviewed papers. Evidence sources span human clinical case reports, mouse model-organism studies, and in vitro mechanistic work, as annotated above.
